

First Lab: Introduction to Microsoft[®] Excel[®]

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1 About the lab session

- There are 10 lab assignments assigned this semester, each counts as 2 points, contributing to 20% of your final course grade.
- If you come to the lab and submit your assignment on time, it's not difficult for you to do well on lab assignments.
- Make sure to come to each lab every Monday from 12:20 to 1:10 p.m. We will start and end each lab on time.
- We will record each lab session mainly for students to review. However, these recordings should not be used as a reason for not attending the lab.
- Every lab assignment is due on Wednesday at 11:59 p.m. You have to submit your Excel file to NTU COOL.

2 Goals for today

- Create an Excel spreadsheet
- Format column or row text and titles
- Add, subtract, multiply, and divide numbers in two or more cells
- Add or delete columns/rows
- Sort your data

3 Introduction

Microsoft Excel is a spreadsheet program used for:

- Data entry and organization

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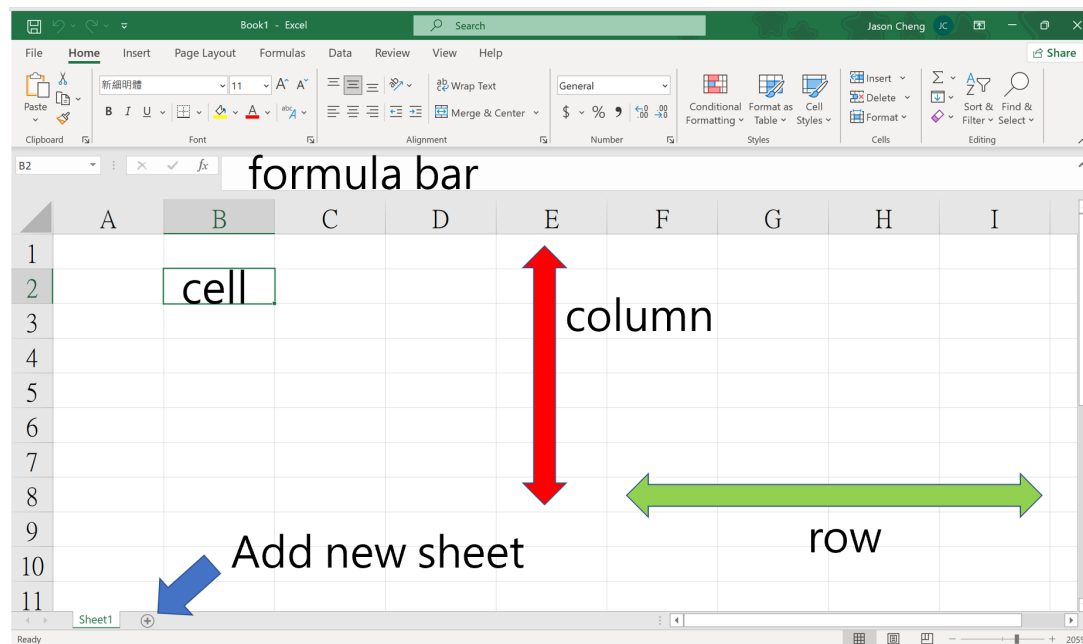
- Automatic calculations and formulas
- Chart creation and data visualization
- Report and table generation

Common applications:

- Budget and expense tracking
- Grade management
- Sales and inventory statistics
- Data analysis and visualization

4 Interface Overview

When you open Excel, you will see:



Area	Function
Ribbon	Located at the top; contains multiple tabs (Home, Insert, Formulas, etc.) with tools and commands.
Name Box	Shows or allows you to type a cell address (e.g., A1).
Formula Bar	Displays or edits the contents/formulas of the selected cell.
Worksheet Tabs	Located at the bottom; used to switch between sheets.
Cells	Intersection of rows (numbers) and columns (letters) where you input data.

5 Entering and Editing Data

- To edit:
 - Select a cell and type new content (or to overwrite the old content).
 - Select a cell and edit in the Formula Bar.

6 Common Functions and Formulas

All formulas start with =:

Formula	Purpose
=A1+A2	Addition
=A1-A2	Subtraction
=A1*A2	Multiplication
=A1/A2	Division
=SUM(A1:A5)	Sum over a range
=AVERAGE(A1:A5)	Average (Mean) of A1 to A5 values
=MAX(A1:A5)	Maximum of A1 to A5 values
=MIN(A1:A5)	Minimum of A1 to A5 values

7 Sorting and Filtering

- Sorting: Data → Sort (Ascending/Descending)
- Filtering: Data → Filter (Display specific data based on conditions)

8 Charts

1. Select the data range.
2. Insert → Choose chart type (Column, Line, etc.).
3. Edit chart title, colors, and labels in Chart Tools.

9 Tips and Shortcuts

Below are useful tips and keyboard shortcuts to work more efficiently in Excel:

Editing and Formatting

- Ctrl + C —Copy.
- Ctrl + X —Cut.
- Ctrl + V —Paste.
- Ctrl + Z —Undo.

- Ctrl + B —Bold.
- Ctrl + I —Italic.
- Ctrl + U —Underline.

Working with Formulas

- = —All formulas start with the equals sign.
- \$ —Toggle absolute/relative references (e.g., A1 → \$A\$1).

Sheet and Workbook Management

- Ctrl + N —New workbook.
- Ctrl + O —Open workbook.
- Ctrl + S —Save workbook.

Other Useful Shortcuts

- Ctrl + F —Find.
- Ctrl + H —Find and replace.

10 Lab assignment

1. Open a new Excel workbook.
2. In the first row, construct obs. and three variables (Age, Gender, GPA).
3. Enter data for 5 hypothetical students (Obs.1 to 5). You make up data by yourself, but ensure that the values are reasonable.
4. Apply a filter to the GPA variable, and sort the data in descending(遞減) order.

Notice

- The file name of your completed lab assignment#1 should be styled as the following format. **"Lab1.(last name)_(first name initials).(date)"** , such as, Lab1.Cheng_HT.9-10-2025.
- Upload your Excel assignment file to NTU COOL before 11:59 pm, September 10th (W), 2025.

11 Reminders (IMPORTANT!!)

- If you would like to **purchase a copy of Howell (2017)**, please complete the **survey** before 3 pm, September 8th (M), 2025.



- Please complete the **Questionnaire** of Students in Psychological Statistics(I) before 5 pm, September 8th (M), 2025.

